



Development and implementation of a parallel ankle rehabilitation robot:
PRANK

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PROPOSAL SUMMARY SHEET

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Title	Development and implementation of a parallel ankle rehabilitation robot: PRANK
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Table 1: Proposal Summary Table

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Abstract

Human gait is a fundamental component of mobility, autonomy, and overall quality of life. Among the joints involved in locomotion, the ankle plays a crucial role in absorbing shock during heel strike and facilitating propulsion during the push-off phase. Impairments in ankle function — whether due to neurological conditions such as stroke, orthopaedic trauma, or post-surgical recovery — can significantly compromise gait, balance, and independence. Robotic rehabilitation has emerged as a promising alternative to traditional therapy, offering consistent, repeatable, and quantitatively measurable support.

This project proposes the design and computational simulation of PRANK (Parallel Robot for ANKle rehabilitation), a three-degree-of-freedom parallel robotic platform actuated by an RRR electric motor configuration and controlled via an admittance algorithm based on real-time force feedback. The work encompasses the complete mechanical design and CAD modelling in SolidWorks with finite element analysis and kinematic workspace verification; the full electronic architecture specification — including motor drivers, a six-axis force/torque sensor, and a dual-core STM32 embedded controller; and the implementation and validation of the admittance control firmware through hardware-in-the-loop testing, in which a physical STM32 development board runs the production firmware while exchanging sensor and actuator data with a ROS 2/Gazebo computational simulation acting as the virtual plant.

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1. PROJECT JUSTIFICATION

Gait is a fundamental aspect of human independence and well-being. Within the gait kinetics, the ankle joint plays a pivotal role in absorbing impact forces during heel strike and generating propulsion during toe-off. When the ankle’s function is impaired—due to conditions such as stroke, orthopedic trauma, or neurodegenerative disease—patients face significant limitations in mobility, balance, and daily autonomy.

Stroke remains one of the leading causes of death and disability worldwide, with approximately 15 million new cases reported each year [1]. The World Stroke Organization emphasizes that this burden continues to rise, particularly in low- and middle-income countries, where access to prevention and treatment is limited. In Colombia, stroke accounts for approximately 120,000 cases and 30,000 deaths annually [2], representing a significant public health challenge that demands effective and accessible rehabilitation strategies.

One of the most debilitating consequences of stroke and other neurological conditions is foot drop—a symptom rather than a disease in itself, arising from multiple underlying pathologies. The most common cause is injury or compression of the peroneal nerve, often linked to trauma, fractures, or prolonged external pressure. Lumbar radiculopathy, particularly involving the L5 nerve root, is another frequent source, typically due to herniated discs or spinal degeneration. Neurological disorders such as stroke, multiple sclerosis, cerebral palsy, and amyotrophic lateral sclerosis can impair motor pathways and lead to foot drop. Muscle-related conditions, including muscular dystrophy and inflammatory myopathies, may also weaken the dorsiflexor muscles. Additionally, systemic diseases like diabetes mellitus contribute through peripheral neuropathy, while tumors, cysts, or autoimmune disorders can cause nerve compression or inflammation. Overall, foot drop reflects a diverse set of neurological, muscular, and systemic pathologies that require careful diagnostic evaluation [3], [4], [5], [6], [7]. Table 1.1 summarizes the epidemiological burden of its primary causes at both the global and national levels, illustrating the scale of the problem and the gaps in local data that motivate targeted intervention.

Cause	Global Data	Colombia Data
Stroke (ACV)	15M cases annually; 5M disabled; 5M deaths	120,000 cases/year; 30,000 deaths
Diabetes-related neuropathy	537M adults with diabetes; 30–50% neuropathy	8–9% adult prevalence; neuropathy common
Peroneal nerve injury	Up to 15% of tibia/knee fractures	Reported in trauma hospitals; no national stats
Neurological disorders (MS, ALS, CP)	MS: 2.8M cases; ALS: 1–2/100,000 incidence	Limited data; hundreds of cases in centers
Muscular disorders (dystrophy, myopathies)	Rare; documented globally in thousands	Very limited data; rare case reports

Table 1.1: Global and Colombian Statistics Related to Foot Drop [1], [2], [8], [9], [10]

Foot drop affects approximately 20% of hemiplegic stroke survivors [11], [12], making it

one of the most prevalent gait impairments in the post-stroke population. Extrapolating from Colombia’s estimated 120,000 annual stroke cases [2], this proportion implies up to **24,000 new foot drop cases per year** from stroke alone — a figure that does not account for cases arising from diabetes, nerve injury, or neurodegenerative disorders. At the global level, the cumulative burden across all aetiologies is substantially larger, yet epidemiological data specific to foot drop remain fragmented and largely absent from low- and middle-income country registries, underscoring the need for accessible, evidence-based rehabilitation technologies in these settings.

Given the scale and diversity of conditions that cause foot drop, effective rehabilitation of the ankle joint is a critical clinical need. However, conventional methods often lack the consistency, objectivity, and sustained engagement required for optimal recovery. Furthermore, existing robotic devices for gait rehabilitation typically do not include mechanisms specifically designed for targeted ankle strength recovery. In this context, robotic rehabilitation systems offer the advantages of repeatability, precise measurement, and the ability to adapt therapy to an individual’s progress. Additionally, mixed reality and serious game interfaces have shown considerable promise in enhancing motivation and participation — key factors that directly influence rehabilitation outcomes [13].

This project addresses these needs through the development of PRANK (Parallel Robot for ANKle rehabilitation), a parallel robotic platform actuated by an RRR configuration and controlled through admittance strategies. It aims to offer a safe, adaptive, and engaging therapeutic environment. In its first phase — the present preproject — the work focuses on ideation, complete mechanical and electronic design, admittance control implementation on an STM32 embedded platform, and full-system validation through a computational simulation with hardware-in-the-loop testing using ROS 2 and a physics simulation environment. Physical fabrication, electronics assembly, mixed reality serious game integration, and clinical pilot trials are planned for Phase 2. PRANK’s long-term design includes **multimodal biofeedback** — visual screen alerts, spatial audio cues, and vibrotactile feedback — to accelerate motor relearning in neurological rehabilitation [13], and ROM validation against an optoelectronic motion capture reference to support its use as a reliable biomechanical assessment tool in future clinical settings.

The present preproject constitutes the first of four planned research phases that together define a doctoral research programme aimed at bringing PRANK from laboratory prototype to a globally deployable rehabilitation platform, as illustrated in Figure 1.1.

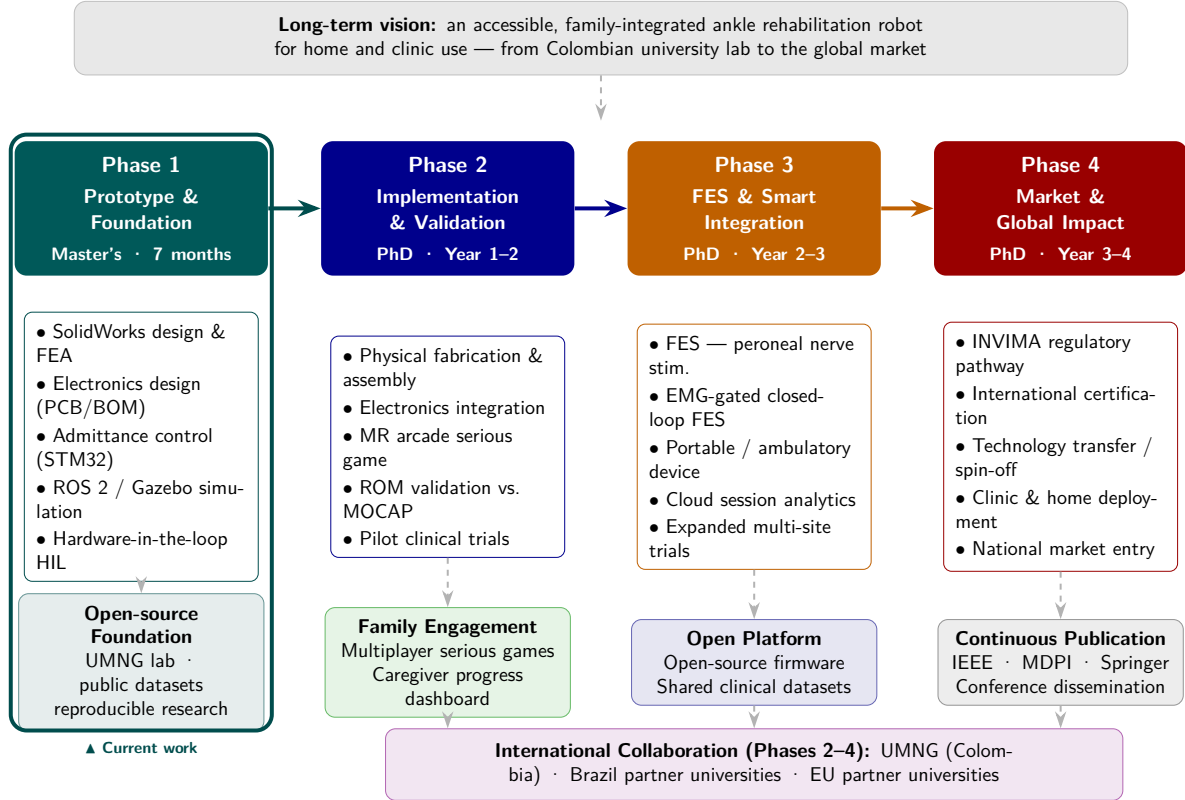


Figure 1.1: Long-term doctoral research roadmap for PRANK. Phase 1 (highlighted, current preproject) covers ideation, mechanical and electronic design, admittance control implementation, and hardware-in-the-loop validation via ROS 2/Gazebo simulation. Phase 2 transitions to physical fabrication, electronics integration, MR serious game development, and clinical pilot trials. Subsequent phases add FES neuromodulation, EMG-based adaptive control, and global market deployment. International collaboration spans Phases 2-4.

2. STATE OF THE ART

The ankle joint complex is one of the most mechanically demanding structures in the human body. Composed of the tibiotalar, talocalcaneal, and transverse-tarsal joints, it enables three rotational degrees of freedom — dorsiflexion/plantarflexion, inversion/eversion, and adduction/abduction — while simultaneously bearing the full body weight during locomotion [14]. As Morris described as early as 1977, the ankle serves simultaneously as a shock absorber at heel contact, a rigid propulsive lever at push-off, and a mechanism for absorbing limb rotation throughout the stance phase [15]. Disruption of any of these functions causes compensatory gait deviations that increase metabolic cost, reduce stability, and elevate fall risk. Lin et al. quantified these effects in 68 post-stroke patients, demonstrating that dorsiflexor muscle strength is the single strongest predictor of gait velocity ($R^2 = 0.30$) and temporal symmetry ($R^2 = 0.36$), while dynamic plantarflexor spasticity drives spatial asymmetry [16]. Normal ranges of motion span approximately $20\text{--}30^\circ$ dorsiflexion, $40\text{--}50^\circ$ plantarflexion, up to 40° inversion, and 15° eversion — values that directly constrain the workspace requirements of any ankle rehabilitation device.

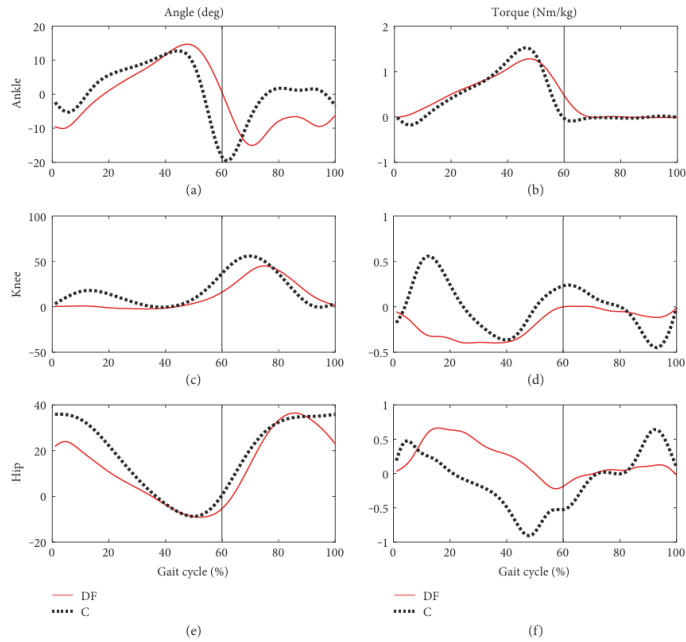


Figure 2.1: Effect of foot drop on spatiotemporal, kinematic, and kinetic parameters during gait. Adapted from [17].

Among the conditions that compromise ankle function, foot drop is one of the most debilitating. It is the inability to perform adequate dorsiflexion during the swing phase of gait, arising from upper motor neurone lesions — most commonly stroke, multiple sclerosis, or spinal cord injury above T12 — and affecting approximately 20% of stroke survivors with lasting consequences [18], [19]. The resulting compensatory strategies — circumduction, hip-

hitching, and steppage gait — are energetically costly and fall-promoting, as illustrated by the kinematic and kinetic deviations documented in Figure 2.1 [17]. Conventional management relies on ankle-foot orthoses (AFOs) and functional electrical stimulation (FES), both of which are primarily orthotic rather than rehabilitative. A meta-analysis of eight randomised controlled trials comparing FES and AFO found equivalent improvements in walking speed for both devices, but identified a critical lack of evidence on kinematic and kinetic outcomes and on whether device-induced gains transfer to everyday settings [20], [21]. This gap motivates the development of active robotic systems capable of delivering repeatable, quantitatively measurable, and adaptive therapy. A long-term study of the NESS L300 FES neuroprosthesis in 16 individuals with chronic hemiparesis demonstrated that consistent use over one year improved 10 m walk velocity from 0.67 m/s to 1.06 m/s, with a 23.8% carry-over effect without the device — evidence that sustained ankle assistance can induce neuroplastic changes [18]. A representative example of this approach is the hybrid FES-robotic system proposed by Kobravi et al., which combines functional electrical stimulation of the peroneal nerve with a robotic assist to restore dorsiflexion in chronic stroke patients, reporting improvements in Fugl-Meyer Lower Extremity (FMA-LE) scores [22] (Figure 2.2).

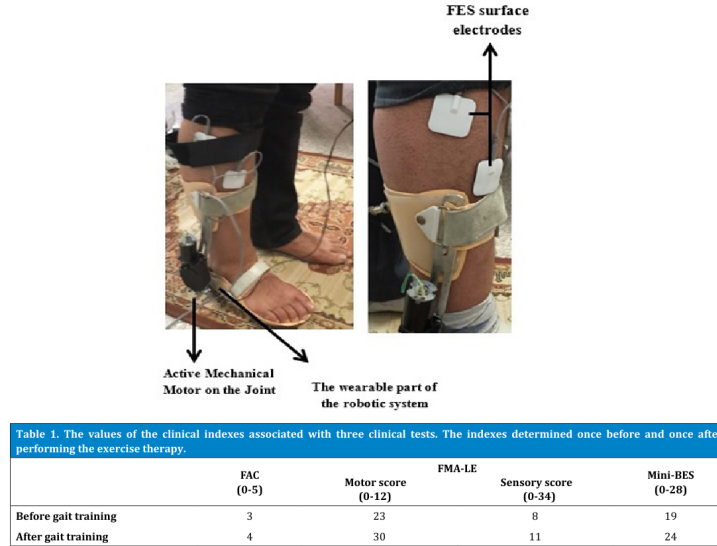


Figure 2.2: Hybrid FES-robotic system for foot drop rehabilitation after stroke and clinical outcome metrics (FMA-LE). Adapted from [22].

Robotic systems have emerged as the most promising platform for delivering this type of therapy consistently, precisely, and adaptively. Among the architectures explored, parallel robots stand out for their rigidity, high load capacity, and ability to produce multi-degree-of-freedom torques simultaneously — properties that align naturally with the complex kinematics of the ankle [23], [24], [25], [26], [27], [28]. The earliest and most influential series of parallel ankle rehabilitation robots was developed by Jamwal and colleagues at the University of Auckland, beginning with a soft parallel robot (SPR) driven by four pneumatic air muscles (Figure 2.3) [23]. Workspace analysis based on Jacobian singularity criteria and a global conditioning number was performed, and a modified genetic algorithm was used to achieve near-unity kinematic isotropy across the feasible workspace. Subsequent work ad-

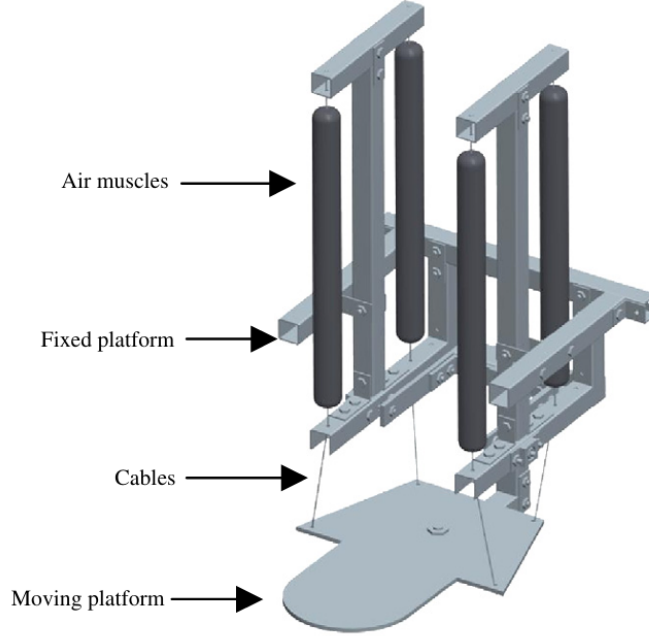


Figure 2.3: Soft parallel robot (SPR) driven by pneumatic air muscles. Adapted from [23].

addressed the forward kinematics problem using a modified fuzzy inference system that outperformed Newton–Raphson iteration in computation speed, making real-time task-space control feasible [23]. A three-stage multi-criteria optimisation framework later decomposed the design problem into kinematic, actuation, and structural stages, applying NSGA-II to simultaneously balance six competing objectives including workspace isotropy, cable tension, and structural compliance [24]. This methodology was further refined with a fuzzy-dominated sorting evolutionary algorithm that improved discrimination among Pareto solutions, providing a reusable template for the dimensional synthesis of future ankle robots [25].

In parallel, Tsoi and Xie designed a three-DoF redundantly actuated platform and demonstrated through singularity analysis that redundant actuation is necessary to avoid singular configurations within the clinical ankle range of motion [14]. Wang et al. extended this direction by performing kinematic simulation of a parallel ankle robot and showing that three-input actuation can cover dorsiflexion $0\text{--}30^\circ$, plantarflexion $0\text{--}50^\circ$, inversion/eversion up to 40° , and adduction/abduction up to 30° [26]. Optimisation of actuator sizing was addressed using a modified differential evolution algorithm with a novel redundant-actuation torque index, outperforming competing methods in global search capability [26]. Spherical parallel mechanisms based on 3-RRS chains have also been explored as an alternative topology: Du et al. obtained eight inverse kinematics solutions analytically and demonstrated accommodation of most patients through an adjustable expansion device [27], while Zuo et al. validated a 3-RRS prototype achieving maximum trajectory errors of 2.5 mm and angular errors within 2° [28]. Figure 2.4 illustrates representative examples of these parallel ankle robot topologies.

A comprehensive review by Miao et al. of platform-based ankle rehabilitation robots from 1980 to 2016 concluded that alignment of the robot’s rotation centre with the anatomical ankle joint centre, appropriate workspace, and reconfigurability are the three most critical

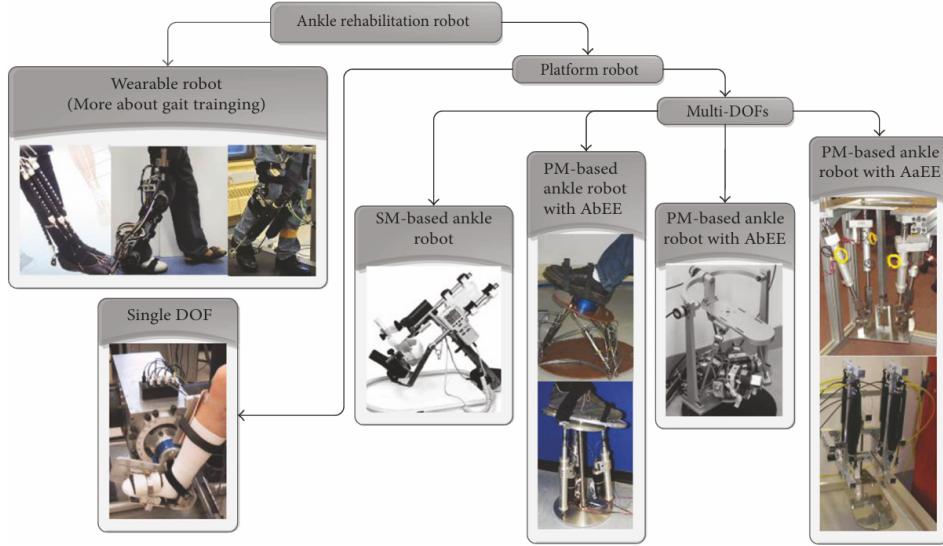


Figure 2.4: Representative parallel ankle rehabilitation robot topologies. Adapted from [29].

design requirements — and that parallel mechanisms best satisfy all three [29]. Li et al. subsequently systematised this finding across 65 articles covering 16 unique parallel ankle robots, identifying rotation-centre alignment, three-DoF coverage, and compliance control with variable parameters as the most pressing open challenges, and noting that clinical trials remain scarce [30] (Figure 2.5). More recently, a 2012–2023 systematic review by Abarca and Elias covering 93 papers confirmed that technology readiness levels remain predominantly at prototype stage and that clinical evidence is limited, underscoring the need for rigorous human-subjects testing [31]. A 2025 review of state-of-the-art ankle rehabilitation robotic devices compared parallel, serial, and exoskeleton topologies across design, actuation, and control dimensions, positioning parallel platforms as the most capable architecture for multi-DoF targeted therapy

The control of ankle rehabilitation robots has evolved from simple position tracking to biologically inspired compliance strategies that actively recruit patient effort. Impedance control was among the first paradigms applied to parallel ankle platforms: Tsoi and Xie used it to regulate both force and motion simultaneously on their redundantly actuated device, exploiting the redundant DoF to control the vertical ankle reaction force — a safety-critical quantity [14]. Jamwal et al. later formalised a four-mode impedance control framework (position control, zero-impedance, and two non-zero impedance levels) and demonstrated through experiments with ten neurologically intact subjects that increasing robotic compliance promotes greater active patient participation, a key neuroplasticity-promoting principle [25]. This was extended to an adaptive version that modifies target impedance in real time based on the patient’s active torque contribution: as participation increases the robot reduces its assistance, and as it decreases the robot provides more support [25]. A parallel and arguably more mature development is the compliant ankle rehabilitation robot (CARR) by Zhang et al., which combines a joint-space position controller with a high-level admittance controller in task space and implements a variable-parameter admittance mode that adapts the prescribed trajectory to real-time ankle measurements, achieving maximum NRMSD below 5.4% [27] (Figure 2.6). A



Figure 2.5: Classification of parallel ankle rehabilitation robots by topology, actuation, and degrees of freedom. Adapted from [30].

subsequent cascade controller with joint-space force distribution ensured all four pneumatic muscles remained in tension throughout the workspace and achieved NRMSD below 2.3% on an injured human ankle [28]. Li et al. implemented a simpler torque-threshold intent detection strategy on a 2-UPS/RRR platform and confirmed small trajectory errors in passive mode and smooth, responsive motion in active mode, providing a lower-complexity baseline for comparison [32]. Most recently, a 2024 cable-driven parallel ankle robot implemented combined position and admittance control for full-cycle rehabilitation, enabling large-angle multi-DoF coupled training with patient-interaction forces modulating the terminal position in real time

Advanced control strategies such as adaptive impedance control, fuzzy logic, and patient-cooperative control have been developed to enhance engagement and safety across the field [25], [27]. EMG-based intent recognition is increasingly integrated to further personalise therapy: Zaffir et al. compared ANFIS, MLP, LSTM, and combined neural network models for estimating dorsiflexion from only two sEMG channels, finding that LSTM achieved highest accuracy and that adding the average rate of change of EMG consistently improved estimation [33]. These results demonstrate that intent detection from tibialis anterior and gastrocnemius sEMG is feasible with a minimal sensor footprint — enabling a low-cost active control mode complementary to admittance-based therapy. Kitatani et al. added a clinically important nuance: elevated ankle muscle co-activation index during single support in hemiplegic patients was negatively correlated with dorsiflexion angle, plantarflexion moment, and ankle power generation, suggesting that EMG-based co-activation monitoring could serve as a real-time biomarker of paretic instability to guide adaptive parameter adjustment [34]. Cable-driven exoskeletons powered by series elastic actuators offer compliant assistance while enhancing gait symmetry [35], and soft robotic platforms such as T-FLEX integrate modular actuation and flexible interfaces for post-stroke rehabilitation [36]. A soft wearable ankle exoskeleton targeting both dropfoot and excessive inversion during swing was published in 2024, illustrating the complementary roles of wearable ambulatory devices and platform-based clinical robots. In the same year, the AMBLE exoskeleton demonstrated through a nine-week clinical study with chronic hemiparetic stroke patients that adaptive impedance control produces durable improvements in toe clearance and dorsiflexion angular velocity — establishing a direct performance benchmark for PRANK’s future clinical evaluation

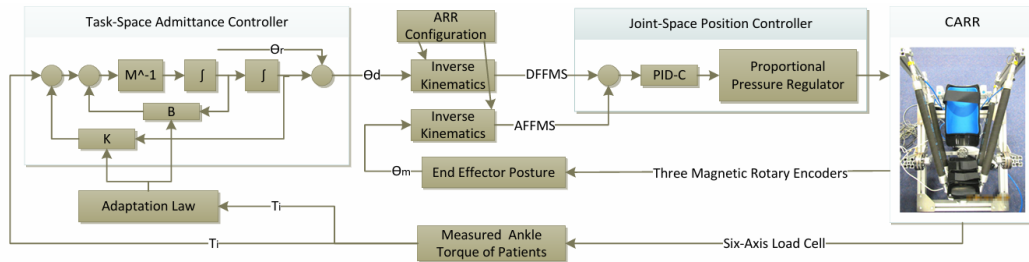


Figure 2.6: Admittance control architecture for ankle rehabilitation robots: force/torque input from the patient is mapped to a motion command through a virtual impedance model. Adapted from [27].

Clinical evidence for robot-assisted ankle therapy, while still growing, increasingly supports its integration into standard rehabilitation protocols. A case report of hybrid FES-robotic training in a chronic stroke patient with foot drop over 16 sessions demonstrated improvements

in the Functional Ambulation Category and Fugl-Meyer Lower Extremity scores, highlighting the cognitive engagement produced when patients voluntarily trigger movement onset [22]. Evaluation of Lokomat gait training with kinematic constraints in 15 hemiparetic patients showed significant improvements in peak knee flexion persisting for at least 20 minutes post-session, suggesting that even short bouts of constrained robotic training produce rapid motor learning [18]. Studies comparing robotic therapy against AFO and FES interventions confirm that robot-assisted systems can match or exceed conventional outcomes in gait restoration and muscle activation [20], [21]. A 2023 multi-DoF reconfigurable parallel ankle robot with mechanically adjustable workspace reported clinical improvements in gait symmetry and muscle activation in post-stroke patients, providing a directly comparable topology and target population benchmark for PRANK. The ROM measurement validation component of PRANK’s design — comparing robot-estimated ankle kinematics against an optoelectronic reference — follows a methodology established across multiple recent studies. Markerless pose estimation systems validated against optoelectronic systems have reported ICC values above 0.91 for limb joints under full ROM tasks [37]. The OpenCap system, validated for lower-extremity kinematics during dynamic tasks, achieved RMSE of 2.0–10.2° relative to marker-based capture, providing context for PRANK’s H3 target of $< 5^\circ$ [38].

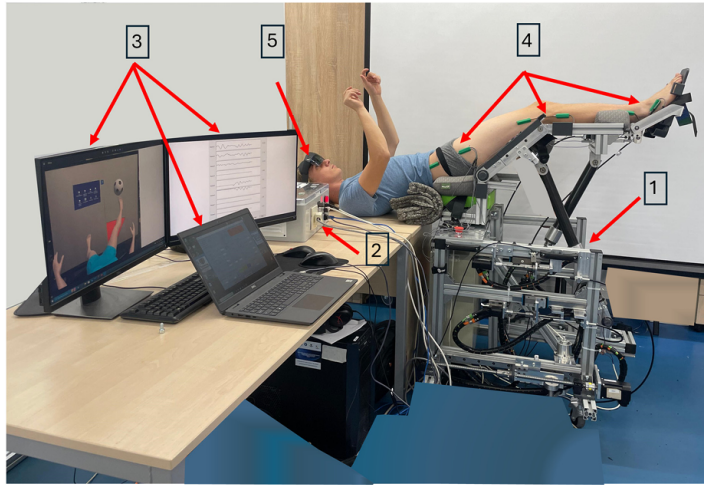


Figure 2.7: Augmented reality and serious gaming integrated with a robotic lower-limb rehabilitation platform. Adapted from [39].

Virtual reality and serious games have progressively emerged as an indispensable complement to robotic hardware in rehabilitation. Systematic reviews highlight their effectiveness in improving postural balance and engagement in patients with neurological disorders [13], and VR-guided active training has been shown to produce superior outcomes to passive trajectory-following through combined tactile and visual feedback that improves ankle strength, flexibility, and balance [32]. Research on hand dominance in VR reach-to-grasp tasks found a significant positive relationship between movement path length and intentional binding — the patient’s implicit sense of agency — for the dominant limb, suggesting that VR rehabilitation interfaces should personalise feedback and difficulty to the affected side to sustain motivation [19]. A 2024 scoping review of serious games for ankle rehabilitation specifically mapped game genres, accessibility requirements, and usability principles, identify-

ing adaptive difficulty scaling and multimodal feedback as the design variables most strongly linked to therapeutic adherence. Augmented reality and serious gaming integrated with a parallel lower-limb robot demonstrated improved ROM outcomes and engagement compared to standard physiotherapy, as illustrated in Figure 2.7, directly supporting the inclusion of a VR serious game as PRANK’s primary active training mode [39]. Integration with remote monitoring platforms is also being explored to improve continuity of care and enable home-based therapy [13], [31].

Despite this progress, several challenges remain open. Ensuring compact, low-cost designs suitable for domestic use; achieving real-time adaptation to individual patient effort and variability; standardising clinical protocols for robot-assisted therapy; and integrating biosignal feedback with cloud-based analytics are all active research frontiers. Portability and modularity have motivated reconfigurable robots that adapt to different patient profiles and therapy stages [31], while the broader trend toward decentralised, patient-centred rehabilitation continues to drive development of systems that combine robotic hardware, intelligent control, and engaging digital interfaces. Future work should focus on scalable architectures, hybrid actuation combining rigid and compliant elements, and intelligent control systems that fuse biomechanical models, machine learning, and human-centred design to close the gap between laboratory prototypes and clinically validated tools.

3. OBJECTIVES

3.1 General

Design and simulate a three-degree-of-freedom parallel ankle rehabilitation robot (PRANK) with an RRR electric motor configuration, developing its mechanical CAD model, electronic architecture design, and admittance-based control system, and validating control performance through a computational simulation environment with hardware-in-the-loop testing on an STM32 development platform.

3.2 Specific Objectives

- Design a three-degree-of-freedom parallel robotic structure actuated by an RRR electric motor configuration, developing a complete 3D CAD model in SolidWorks suitable for structural FEA analysis, kinematic workspace verification, and importation into the computational simulation environment.
- Define the electronic architecture of PRANK — including motor drivers, six-axis force/torque sensing, and dual-core STM32 embedded controller — producing a complete design specification and PCB layout without physical implementation in this phase.
- Implement an admittance-based control system with passive, assistive, and resistive rehabilitation modes, simulate the full robot–patient interaction in a computational environment, and validate controller performance through hardware-in-the-loop testing using a physical STM32 development board as the embedded control target.

4. DELIMITATION AND SCOPE

4.1 Type of Research

This project is classified as **applied** research with an **experimental simulation** component, since it involves the design, modelling, and controlled computational simulation of a physical system — the PRANK parallel ankle rehabilitation robot — under conditions defined by the investigator. It also has a **descriptive** component, as one of its core objectives is to characterise the control performance of the admittance system through simulation-based experiments and hardware-in-the-loop testing. The experimental phase will determine whether the system meets the performance thresholds stated in the project hypotheses (H1, H2, H3), while the descriptive phase will document the system’s behaviour across its three rehabilitation modes.

4.2 Delimitation Conditions

Space

All design, software development, simulation, and hardware-in-the-loop testing activities will be carried out at the facilities of the **Universidad Militar Nueva Granada (UMNG)**, Bogotá, Colombia. Specifically, the mechatronics and robotics laboratories of the Faculty of Engineering will be used for CAD design, embedded firmware development, and HIL validation with the STM32 development board.

Time

The project will be executed during **2026**, with a planned duration of seven (7) months. The schedule follows the Gantt chart presented in Chapter 8, covering the phases of mechanical design and CAD modelling, electronic architecture specification, embedded control implementation, and hardware-in-the-loop validation.

Universe

The validation experiments (H2 and H3) are conducted entirely in the computational simulation and hardware-in-the-loop environment; no human subjects are required in this phase. The simulation stimuli (joint reference trajectories and interaction force profiles) are derived from biomechanical data reported in the literature for healthy adult ankle motion. Human subjects experiments are planned for Phase 2, following physical fabrication and institutional ethics review.

Scope

The project encompasses:

- **Mechanical design:** Design of a three-degree-of-freedom parallel robotic structure with an RRR electric motor configuration in SolidWorks, including FEA, kinematic

workspace verification, and URDF/SDF export for simulation. Physical fabrication is deferred to Phase 2.

- **Electronic architecture:** Specification of the complete electronics — brushless DC motors with encoders, six-axis load cell, motor drivers (CAN bus), and dual-core STM32 embedded platform — in the form of schematic, PCB layout, and bill of materials. Physical board assembly is deferred to Phase 2.
- **Embedded control:** Implementation and hardware-in-the-loop validation of an admittance-based control system in C/C++ (FreeRTOS on STM32), supporting passive, assistive, and resistive rehabilitation modes (H2).
- **Computational simulation:** ROS 2/Gazebo simulation environment using the URDF model as the virtual plant, providing virtual sensor readings to the STM32 HIL board and receiving motor commands to close the physics loop (H3).

The project does **not** include in this phase:

- Physical fabrication or assembly of the robot or custom electronics.
- Mixed reality or serious game interface development (planned for Phase 3).
- Clinical trials or human subjects experiments.
- ROM validation against an optoelectronic motion capture system (planned for Phase 2).
- Assessment of therapeutic efficacy or rehabilitation outcomes.
- Commercialisation or regulatory approval activities.

4.3 Research Variables

4.3.1 Independent Variables

Independent variables are those controlled or manipulated by the investigator during the simulation and HIL experiments:

- **Control mode:** The rehabilitation mode selected for each trial — passive, assistive, or resistive. This variable determines the admittance parameter configuration and the role of the simulated patient interaction forces.
- **Admittance parameters (K_d , B_d , M_d):** The virtual stiffness, damping, and inertia matrices of the admittance controller (Equation 6.3). These are varied systematically to characterise controller behaviour and tune performance.
- **Prescribed trajectory / ROM target:** The reference ankle trajectory defined for each simulation experiment (e.g., sinusoidal dorsiflexion/plantarflexion at a specified amplitude and frequency).
- **Simulation mode:** Whether the experiment runs in pure-software simulation (ROS 2/Gazebo only) or HIL mode (STM32 firmware + ROS 2/Gazebo plant), used to quantify embedded implementation fidelity (H3).

4.3.2 Dependent Variables

Dependent variables are those measured as outputs of the simulation or HIL system:

- **Ankle range of motion (ROM):** The angular excursion of the simulated ankle in dorsiflexion/plantarflexion, inversion/eversion, and internal/external rotation. Primary variable for H1 workspace verification.
- **Interaction force and torque:** The six-component wrench injected into the simulation as the virtual patient effort input. Input to the admittance controller and primary observable for evaluating H2.
- **Controller tracking error (RMSE):** Root mean square deviation between prescribed reference trajectory and simulated joint trajectory, in degrees. Used to quantify control performance (H2).
- **Controller settling time:** Time required for the system to reach steady-state after a step reference change, in milliseconds (H2).
- **HIL trajectory deviation:** Root mean square deviation between HIL and pure-software simulation trajectories under identical inputs, in degrees. Primary metric for H3.
- **Communication latency:** USB serial round-trip time between the ROS 2 host and STM32, in milliseconds. Confirms real-time operation readiness (H3).

4.4 Limitations

The following limitations are acknowledged and will be addressed in the interpretation of results:

- **Simulation fidelity:** The Gazebo physics model approximates the robot’s dynamics based on the SolidWorks CAD parameters. Unmodelled effects — joint friction, flexibility, sensor noise — will be characterised and bounded but may affect generalisability to the physical prototype.
- **No physical hardware validation:** All controller performance results are obtained in simulation and HIL. Differences between virtual and physical behaviour (actuator nonlinearities, real cable routing, thermal effects) will only be quantified after Phase 2 fabrication.
- **Simulated patient forces:** Patient interaction forces are generated from literature-derived biomechanical profiles rather than real muscle activity, which may not capture the full variability of clinical populations.
- **Short time horizon:** The seven-month duration precludes longitudinal assessment or iterative design refinement beyond the initial simulation campaign.
- **HIL communication throughput:** USB serial bandwidth sets an upper bound on HIL loop rate; sufficiency for real-time control will be verified experimentally as part of H3.

5. HYPOTHESIS

H1 — Mechanical Design: The PRANK parallel robotic structure, as defined by its SolidWorks CAD model and kinematic analysis, will achieve a functional workspace covering at least $\pm 25^\circ$ of dorsiflexion/plantarflexion, $\pm 15^\circ$ of inversion/eversion, and $\pm 10^\circ$ of internal/external rotation — values established as the minimum therapeutic range of motion required for ankle rehabilitation [14], [40].

H2 — Control System: The admittance-based control system implemented in the computational simulation environment will demonstrate stable and responsive behaviour across passive, assistive, and resistive rehabilitation modes, with a settling time under 500 ms and a force tracking error below 15% of the commanded torque, ensuring safe and adaptive interaction with the patient throughout therapy [27], [32].

H3 — Hardware-in-the-Loop Validation: The hardware-in-the-loop configuration — in which the STM32 development board executes the admittance control firmware in real time while exchanging state and command data with the computational simulator — will reproduce the control trajectories obtained in the pure software simulation with a root mean square deviation below 2° and a communication latency compatible with real-time operation (< 10 ms cycle), demonstrating the feasibility of the embedded implementation before physical fabrication.

6. THEORETICAL FRAMEWORK

6.1 Human Gait

Human gait is one of the most important tasks of daily life, as it allows people to maintain independence and live with autonomy, mobility, and dignity. Impairments in gait due to neurological or musculoskeletal conditions can significantly affect quality of life, making rehabilitation a critical component of recovery and functional reintegration.

The gait analysis is carried out on the basis of the gait cycle, which can be taken as a representation of a person's walking patterns; comparison of several cycles indicates the variability of the pattern [18]. The gait cycle is normalized from 0% to 100% because each individual pattern has different absolute timing, making direct cycle-length comparisons non-objective [19]. It can be described with spatial and temporal parameters.

6.1.1 Spatial parameters

The main spatial parameters are shown in Figure 6.1 and described below:

- **Step:** the movement of one foot in front of the other.
- **Stride:** a step of one foot followed by a step of the other foot; one complete gait cycle.
- **Foot contact:** the beginning of the gait cycle, referred to as *heel strike* in healthy individuals.
- **Step length:** distance traveled by one foot in front of the same reference point of the other foot.
- **Stride length:** distance between two consecutive heel strikes of the same foot.
- **Step width:** mediolateral separation of the feet, also known as stride width.

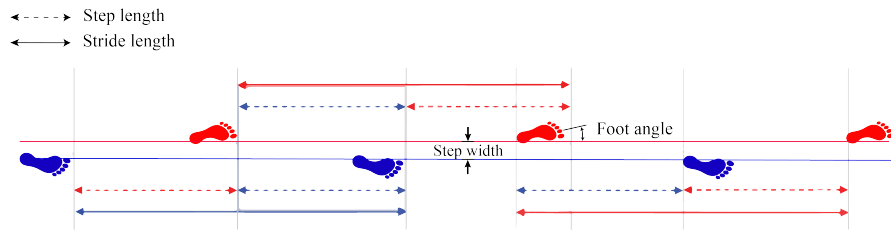


Figure 6.1: Gait spatial parameters. Source: edited by author from [18].

6.1.2 Temporal parameters

Temporal parameters are listed below:

- **Stride time:** duration of one gait cycle (time between two consecutive foot strikes of the same foot).
- **Cadence:** a commonly used indirect measure of cycle frequency, expressed as:

$$\text{Cadence} = \frac{\text{number of cycles}}{\text{time interval}} \quad (6.1)$$

- **Walking speed:** distance traveled per unit time in m/s. If cadence is in steps per minute and stride length is in meters:

$$v = \frac{\text{cadence} \times \text{stride length}}{120} \quad (6.2)$$

Gait is globally divided into two phases: *stance* and *swing*. Stance occurs when the foot is in contact with the ground; swing occurs when it is not. The stance phase ends at toe-off, which is simultaneously the onset of swing, as illustrated in Figure 6.2.

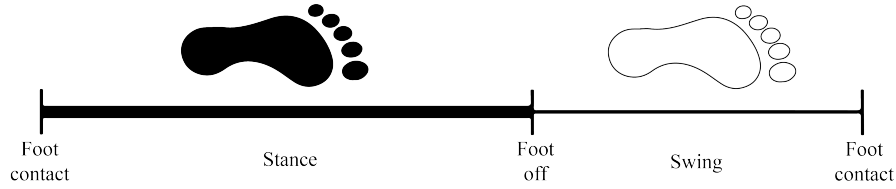


Figure 6.2: One side gait phases. Source: edited by author from [18].

Including both legs, the gait cycle is divided into first double support (both feet in contact), single support, second double support, and swing, as shown in Figure 6.3. Single support and swing are long phases, so each is further subdivided into *early*, *middle*, and *late* sub-phases [18], [41].

Any impairment in the ankle joint manifests as alterations in gait parameters, disrupting the cyclical and repetitive nature of walking. This leads to increased energy expenditure and triggers compensatory mechanisms in other muscle groups, often resulting in instability and a heightened risk of falling.

6.2 Anatomy of the Ankle Joint Complex

The ankle joint complex (AJC) is not a simple hinge joint but a mechanically sophisticated structure composed of three articulations working in concert: the **tibiotalar (talocrural)**

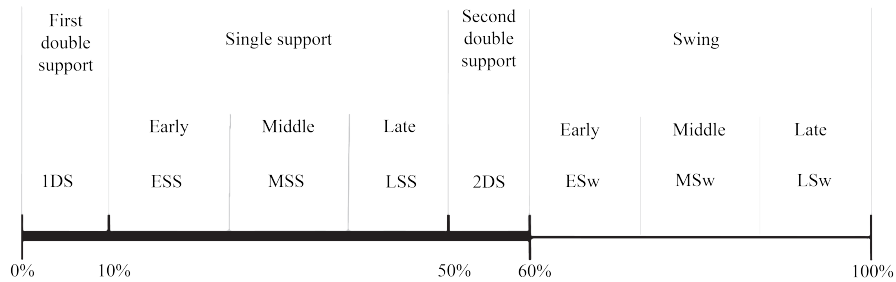


Figure 6.3: Both sides gait phases. Source: edited by author from [18].

joint, the **talocalcaneal (subtalar) joint**, and the **transverse-tarsal (talocalcaneonavicular) joint** [40]. Together, these joints enable three rotational degrees of freedom that are essential for locomotion.

6.2.1 Tibiotalar (Talocrural) Joint

The talocrural joint is formed by the distal tibia, the distal fibula, and the trochlea of the talus. The medial and lateral malleoli form a mortise that constrains the talus and confers primary stability. Although the joint functions predominantly as a hinge, the conical geometry of the trochlea surface and the oblique orientation of its rotation axis introduce a coupled axial rotation component during dorsiflexion and plantarflexion, making it biomechanically more complex than a pure hinge [40]. The talus is widest anteriorly, so the joint is most congruent and stable in dorsiflexion — a clinically important asymmetry.

Stability is provided by three ligament groups [40], [42]:

- **Lateral collateral ligament complex:** anterior talofibular ligament (ATFL), calcaneofibular ligament (CFL), and posterior talofibular ligament (PTFL). The ATFL is the most frequently injured ligament in ankle sprains.
- **Medial collateral (deltoid) ligament:** a broad, triangular ligament complex that resists eversion and provides medial stability.
- **Tibiofibular syndesmosis:** anterior and posterior tibiofibular ligaments plus the interosseous membrane, maintaining the integrity of the mortise under dynamic loading.

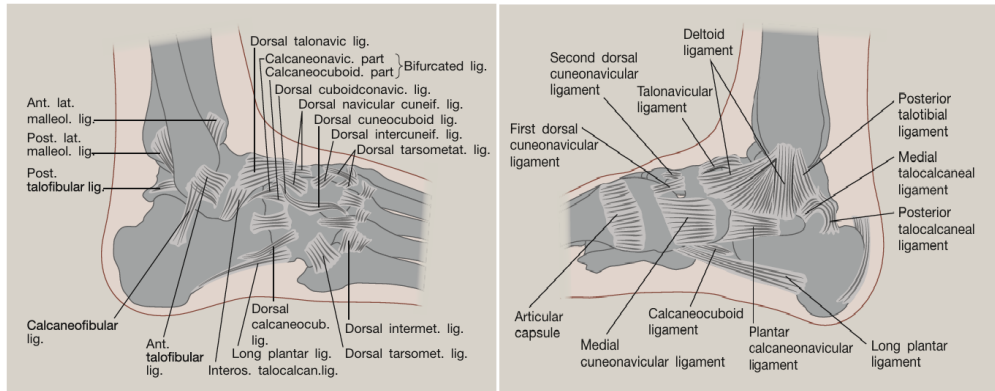


Figure 6.4: Ligament anatomy of the ankle joint complex. Source: author.

6.2.2 Subtalar (Talocalcaneal) Joint

The subtalar joint is formed by three facets between the talus and calcaneus — posterior, middle, and anterior — and functions as a triplanar, uniaxial joint [40]. Its oblique rotation axis (inclined approximately 42° from horizontal and 16° from the sagittal plane) means that motion at this joint is inherently triplanar: dorsiflexion–abduction–eversion in one direction and plantarflexion–adduction–inversion in the other. This coupling is fundamental to gait:

the subtalar joint unlocks during swing to allow shock absorption, and locks during push-off to convert the foot into a rigid lever [40]. The interosseous talocalcaneal ligament is its primary restraint, complemented by the CFL (the only ligament bridging both the talocrural and subtalar joints) and surrounding tendons.

6.2.3 Muscles and Tendons

The principal muscle groups acting on the ankle are [40], [42]:

- **Dorsiflexors:** tibialis anterior (primary), extensor digitorum longus, extensor hallucis longus. These muscles prevent foot drop during swing and control forefoot landing at heel strike.
- **Plantarflexors:** gastrocnemius and soleus (triceps surae, acting via the Achilles tendon), tibialis posterior, peroneus longus and brevis. The triceps surae generates the propulsive push-off force.
- **Invertors / Evertors:** tibialis posterior (inversion), peroneus longus and brevis (eversion), which contribute to frontal-plane stability and foot posture.

The Achilles tendon, transmitting triceps surae forces to the calcaneus, is the largest and strongest tendon in the body. Its role in energy storage and return during the stance-to-swing transition makes it a key structure for walking efficiency [43], [44].

6.3 Kinematics and Kinetics of the Ankle in Gait

6.3.1 Range of Motion

The clinical and biomechanical literature reports the following reference ranges for the AJC [14], [40], [42]:

Table 6.1: Reference ranges of motion of the ankle joint complex

Movement	Full passive ROM	Required for gait
Dorsiflexion	10°–20°	≈ 15°
Plantarflexion	40°–55°	≈ 20°
Inversion	20°–35°	≈ 5°–10°
Eversion	10°–15°	≈ 5°
Internal / External rotation	±5°–15°	Coupled to subtalar motion

During normal walking, the total sagittal-plane excursion is approximately 30° (from ~ 5° dorsiflexion at heel strike to ~ 20° plantarflexion at toe-off). Stair climbing demands up to 37° dorsiflexion and 56° plantarflexion, illustrating the need for wider workspace in rehabilitation devices targeting functional activities beyond level-ground walking. These values directly constrain the workspace requirements of PRANK: the robot must cover at least ±25° dorsiflexion/plantarflexion, ±15° inversion/eversion, and ±10° internal/external rotation to enable comprehensive ankle rehabilitation.

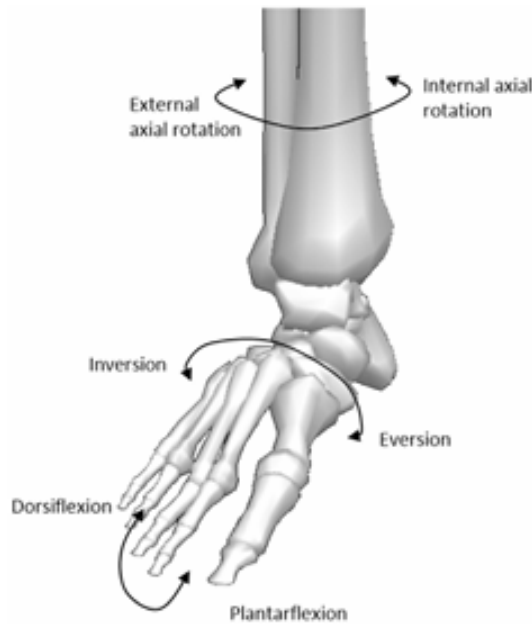


Figure 6.5: Reference planes and angular degrees of freedom of the ankle joint complex. Source: author.

6.3.2 Ankle Kinematics During the Gait Cycle

The ankle follows a characteristic kinematic pattern across the gait cycle [40], [42]:

- **Heel strike (0%):** The foot lands in slight plantarflexion ($\sim 5^\circ$); tibialis anterior activates eccentrically to lower the forefoot and control impact.
- **Loading response (0–12%):** Controlled plantarflexion to $\sim 5^\circ$ as the foot flattens to the ground.
- **Mid-stance (12–50%):** Progressive dorsiflexion as the tibia advances over the fixed foot, reaching peak dorsiflexion of $\sim 10^\circ$ – 15° .
- **Terminal stance and pre-swing (50–62%):** Rapid plantarflexion to $\sim 20^\circ$, with the gastrocnemius–soleus generating the propulsive push-off burst.
- **Swing (62–100%):** Dorsiflexion returns to neutral to clear the ground; failure to achieve this is the defining feature of *foot drop*.

6.3.3 Ankle Kinetics and Power

The kinetics of the ankle during gait are dominated by two events [42], [44]:

- **H1 — controlled plantarflexion (loading response):** a small energy-absorption event as the tibialis anterior decelerates forefoot drop.

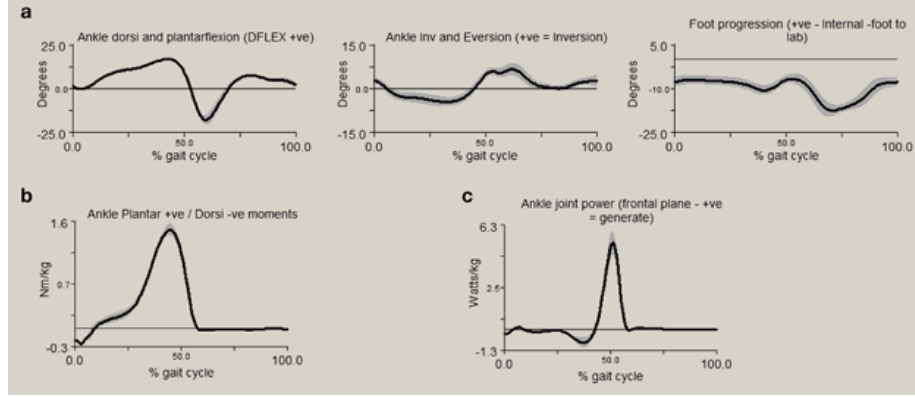


Figure 6.6: Ankle joint kinematics and kinetics during the gait cycle: angular displacement, moment, and power profiles. Source: author.

- **A2 — push-off power burst (pre-swing):** the largest energy-generation event in normal gait. Peak plantarflexion torque reaches **1.4–1.6 Nm/kg** of body mass, and peak ankle power reaches **2.5–4.0 W/kg** at approximately 55–60% of the gait cycle [42], [43].

For a 70 kg adult this corresponds to approximately 98–112 Nm peak torque and 175–280 W peak power at push-off — the highest power output of any lower-limb joint during walking. These values are the primary mechanical specifications for PRANK: the robot’s actuators and structural frame must sustain these loads during resistive and assistive training without compromising safety. Table 6.2 summarises the key kinetic parameters:

Table 6.2: Normative ankle kinetic parameters during level-ground walking (70 kg reference subject)

Parameter	Normalised	Absolute (70 kg)
Peak plantarflexion torque (push-off)	1.4–1.6 Nm/kg	98–112 Nm
Peak ankle power (push-off, A2)	2.5–4.0 W/kg	175–280 W
Peak dorsiflexion torque (swing)	<0.2 Nm/kg	<14 Nm
Peak angular velocity (push-off)	~4.3 rad/s	—

6.4 Biomechanical Modelling of the Ankle

Biomechanical models of the ankle range from simplified single-axis representations to high-fidelity multi-segment finite element models [45], [46]. For robot control purposes, the most relevant level of abstraction is the **rigid-body task-space model**: the ankle is represented as a spherical joint with three rotational degrees of freedom, and the interaction between the robot’s footplate and the patient’s foot is described by a wrench (force and torque vector) applied at a defined reference point.

A key insight from the biomechanics literature is that the ankle’s mechanical behavior is not constant: its **quasi-stiffness** — the slope of the torque-angle curve — varies from

$\sim 2 \text{ Nm}/^\circ$ during controlled plantarflexion (loading response) to $\sim 8 \text{ Nm}/^\circ$ during push-off [43]. This nonlinear, phase-dependent impedance means that fixed-gain position controllers are insufficient for safe human–robot interaction: a controller that is stiff enough to guide passive ROM training would resist voluntary patient effort in active training modes. This is the fundamental motivation for **admittance control**.

6.4.1 Admittance Control and Its Connection to Ankle Biomechanics

Admittance control is a human–robot interaction paradigm in which the robot accepts force or torque inputs from the patient and responds by modifying its motion output [27], [32]. The admittance relationship maps a measured interaction wrench $\mathbf{F}$ (6-dimensional force/torque vector) to a commanded velocity or position deviation $\dot{\mathbf{x}}$ through a virtual mechanical impedance:

$$M_d \ddot{\mathbf{x}} + B_d \dot{\mathbf{x}} + K_d \mathbf{x} = \mathbf{F} \quad (6.3)$$

where M_d , B_d , and K_d are the desired virtual inertia, damping, and stiffness matrices, respectively. By tuning these parameters, the therapist or the control algorithm can select the robot’s apparent mechanical resistance independently of its physical structure.

The connection to ankle biomechanics is direct:

- **Passive training mode:** K_d is set high and B_d is moderate — the robot guides the ankle along a prescribed trajectory, mimicking the feel of manual physiotherapy.
- **Assistive training mode:** K_d is reduced and M_d is low — the robot follows the patient’s voluntary motion with minimal resistance, encouraging active participation. This exploits the neuroplasticity principle that patient-cooperative training produces greater cortical reorganisation than purely passive movement.
- **Resistive training mode:** B_d is increased — the robot adds a velocity-proportional resistive torque that opposes the patient’s movement, providing progressive strength training analogous to the push-off loading experienced in normal gait.

Adaptive admittance control extends this by varying M_d , B_d , K_d in real time based on patient performance metrics — for example, reducing K_d as the patient’s active torque contribution increases [25], [27]. This directly replicates the role of the ankle’s own variable quasi-stiffness in normal gait, making the robot biomechanically compatible with the natural mechanics of the joint it is rehabilitating.

The biomechanical parameters from Table 6.2 provide concrete bounds for the admittance design: the virtual stiffness K_d must be bounded below by the minimum torque needed to passively move the ankle (approximately $0.2 \text{ Nm}/^\circ$ based on cadaveric measurements) and above by the peak torque the device must safely resist ($1.6 \text{ Nm/kg} \times \text{patient mass}$), while the virtual damping B_d must be sufficient to attenuate oscillations within the ankle’s natural bandwidth of approximately 3–5 Hz during gait.

6.5 Motor Rehabilitation and Neuroplasticity

Motor recovery after injury or neurological impairment relies on neuroplasticity — the brain’s ability to reorganize through repetitive, goal-oriented movement. Rehabilitation robotics

leverage this principle by enabling high-intensity, task-specific training with precise, reproducible mechanics. Key motor learning principles include augmented feedback, movement variability, and active participation, all of which enhance cortical reorganisation and functional recovery [47], [48]. PRANK integrates these principles through interactive VR tasks that provide real-time visual and haptic feedback, and through an admittance control architecture that demands and rewards active patient effort.

6.6 Human–Machine Interfaces in Rehabilitation

Effective rehabilitation requires intuitive and responsive interfaces between the user and the robotic system. These interfaces must accommodate physical comfort, cognitive engagement, and safety. Advances in flexible electronics and soft robotics have enabled more ergonomic designs that conform to the user’s anatomy [49]. Virtual reality (VR) environments provide immersive feedback, increasing motivation and facilitating motor learning through gamified rehabilitation tasks [50]. Research shows that combining tactile and visual feedback improves ankle strength, flexibility, and balance beyond what passive trajectory-following achieves, and that personalising the game interface to the affected limb side sustains the patient’s implicit sense of agency [19].

6.7 Clinical Evaluation Metrics

Quantifying rehabilitation outcomes is essential for validating system effectiveness. Clinical metrics include range of motion, gait symmetry, balance, and patient-reported outcomes. Instruments such as the Rehabilitation Measures Database offer standardised tools for assessing ankle functionality and tracking progress over time [51]. In the context of PRANK Phase 1, primary quantitative outcomes are: (1) ankle ROM workspace coverage across all three axes verified in the Gazebo simulation environment; (2) force tracking error and settling time of the admittance controller validated through ROS 2/Gazebo hardware-in-the-loop experiments; and (3) simulation fidelity metrics comparing the virtual model against the analytical kinematic and dynamic model. These metrics are directly linked to the three hypotheses of the project and inform the design of the pilot study protocol.

7. MATERIALS AND METHODS

This chapter describes the system architecture and development methodology of PRANK — a Parallel Robot for ANKle rehabilitation. The project is structured in four sequential stages spanning mechanical design, electronic architecture specification, embedded control implementation, and hardware-in-the-loop (HIL) simulation validation. In the HIL configuration, the STM32 development board runs the production admittance control firmware and interacts in real time with a ROS 2/Gazebo simulation that provides virtual sensor readings (encoders, force/torque) and actuates the simulated robot based on the firmware commands — replacing the physical motors and sensors that will be integrated in Phase 2. This staged structure supports project scheduling and provides clear deliverables and entry/exit criteria at each phase.

Figure 7.1 presents the overall system architecture, showing the integration between the mechanical platform, embedded hardware, and the computational simulation layer.

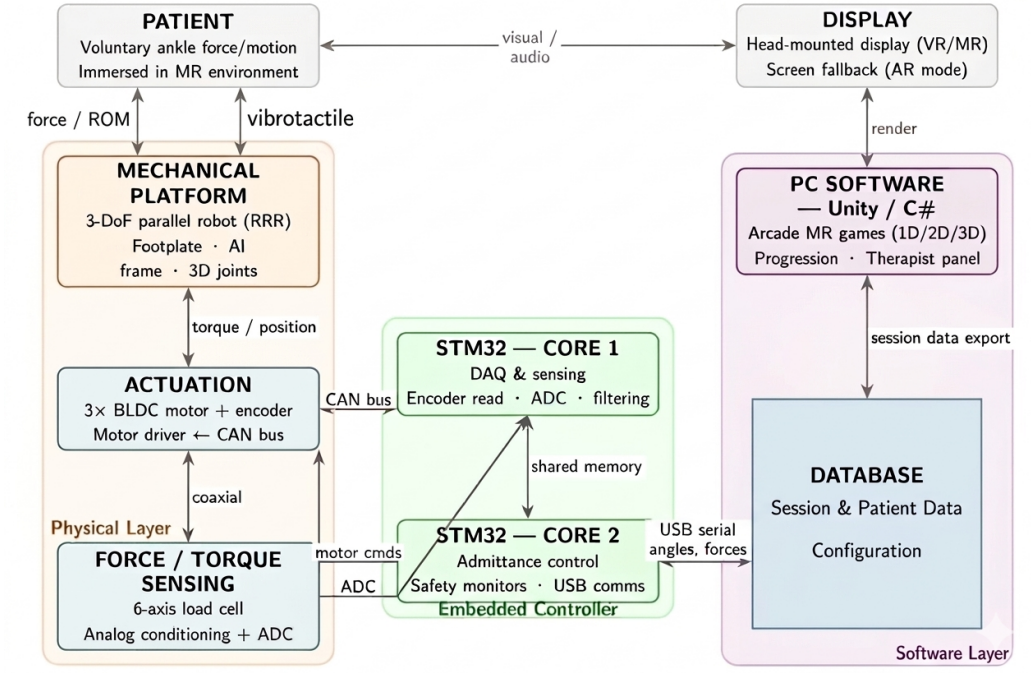


Figure 7.1: PRANK system architecture: mechanical platform, dual-core STM32 embedded controller, and software layer with mixed reality interface.

7.1 Stage 1 — Mechanical Design and CAD Modelling

The PRANK robot is a three-degree-of-freedom (DoF) parallel structure designed to produce controlled rotational motion at the ankle joint in three anatomical planes: dorsiflexion/plantarflexion, inversion/eversion, and internal/external rotation. The kinematic configuration

follows an RRR (revolute–revolute–revolute) chain arrangement, in which three independent motor-driven arms converge on a common moving platform — the footplate — through spherical or universal joints.

The mechanical structure is constructed from lightweight aluminium profiles and 3D-printed structural joints, combining rigidity with modularity and ease of component replacement. The footplate includes an adjustable strapping system and a contoured sole support to accommodate different foot sizes and ensure reliable mechanical coupling between the patient’s foot and the robot. Rotation centre alignment with the anatomical ankle joint centre is a primary design constraint, implemented through geometric calibration of the kinematic chain during the design phase.

The full 3D mechanical design is developed in **SolidWorks**, which is also used for structural simulation (FEA) to verify that the frame, joints, and footplate withstand the peak loads reported in the biomechanics literature (up to ~ 112 Nm at the ankle during push-off equivalent loading). Kinematic workspace coverage — at minimum $\pm 25^\circ$ dorsiflexion/plantarflexion, $\pm 15^\circ$ inversion/eversion, and $\pm 10^\circ$ internal/external rotation (H1) — is verified through SolidWorks Motion analysis. The resulting CAD model is exported in URDF/SDF format for import into the ROS 2/Gazebo simulation environment.

Stage deliverable: Complete SolidWorks CAD model with FEA report, workspace verification results, and URDF/SDF simulation export.

7.2 Stage 2 — Electronic Architecture Design

Each of the three kinematic chains is actuated by a **brushless DC motor** with an integrated incremental encoder, providing position feedback with sufficient resolution for ankle-level kinematics ($< 0.1^\circ$ angular resolution at the footplate). Motor drivers communicate with the embedded controller via a CAN bus, providing a noise-robust, real-time communication channel suitable for multi-axis coordinated motion.

Force and torque sensing at the robot–patient interface is performed by a **six-axis load cell** mounted between the footplate and the last structural link, providing the full interaction wrench ($F_x, F_y, F_z, M_x, M_y, M_z$) required by the admittance controller to detect rotational intent at the ankle. Analog signals from the load cell are conditioned and digitised by a dedicated ADC front-end before transmission to the controller.

The embedded processing platform is a dual-core **STM32** microcontroller:

- **Core 1 — Sensing and data acquisition:** Reads encoder positions, load cell measurements, and auxiliary sensors at a fixed sampling rate; performs signal filtering and packaging; passes processed data to Core 2 via shared memory with hardware semaphore.
- **Core 2 — Control and communications:** Executes the admittance control law, generates motor torque commands, manages safety monitors, and handles serial communication with the ROS 2 simulation host.

In this phase, the electronic design is fully specified — including schematic, PCB layout, and bill of materials — but **physical board fabrication and assembly are deferred to Phase 2**. A commercially available STM32 development board is used in Stage 4 for HIL validation.

Stage deliverable: Complete electronic design package (schematic, PCB layout, BOM) and STM32 firmware architecture ready for HIL deployment.

7.3 Stage 3 — Embedded Control Implementation

The admittance control law and mode-switching logic are implemented in **C/C++** directly on the STM32 platform under **FreeRTOS**. The control architecture implements three rehabilitation modes — passive, assistive, and resistive — all built on the admittance framework described in the theoretical framework (Equation 6.3):

- **Passive mode:** high virtual stiffness K_d guides the ankle through a predefined reference trajectory regardless of patient effort.
- **Assistive mode:** reduced stiffness allows the patient’s voluntary interaction forces to modify the prescribed trajectory, encouraging active participation.
- **Resistive mode:** velocity-proportional virtual damping B_d opposes patient motion, providing progressive mechanical resistance for strength training.

Safety mechanisms implemented in firmware include: per-axis torque saturation, workspace boundary enforcement, watchdog timer, and a software emergency-stop interface.

In the HIL setup, the STM32 firmware receives simulated sensor data from the ROS 2/Gazebo plant (joint angles from virtual encoders and interaction forces from the simulated load cell) over a USB serial link, computes the admittance control output, and returns motor commands that are applied back to the Gazebo simulation. From the firmware’s perspective, the simulation is indistinguishable from real hardware, exercising the full embedded control loop including interrupt timing, FreeRTOS task scheduling, and CAN-bus driver logic.

Figure 7.2 illustrates the HIL simulation architecture.

Stage deliverable: Embedded admittance control firmware running on STM32 development board, integrated with ROS 2/Gazebo HIL loop.

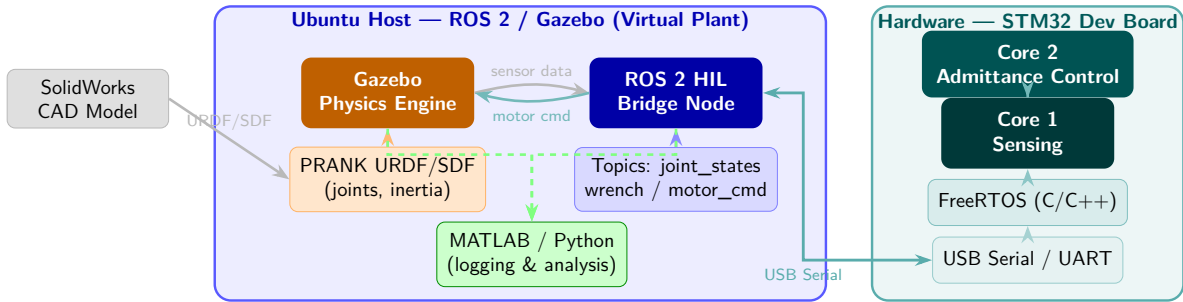


Figure 7.2: Hardware-in-the-loop architecture for PRANK Phase 1. The Ubuntu/ROS 2 host runs Gazebo as the virtual plant, simulating encoder and load cell outputs that are exchanged with the physical STM32 development board via USB serial. The STM32 executes the production admittance control firmware (FreeRTOS, C/C++) and returns motor commands that close the physics loop in Gazebo. The PRANK URDF/SDF model is exported from SolidWorks.

7.4 Stage 4 — HIL Validation and Performance Evaluation

The final stage validates the full HIL loop and evaluates the admittance control system against the project hypotheses.

H1 — Workspace verification: The kinematic model is commanded to its boundary poses in each axis within the SolidWorks Motion and Gazebo simulation environments, and achieved angles are compared against the H1 thresholds ($\pm 25^\circ$, $\pm 15^\circ$, $\pm 10^\circ$).

H2 — Controller performance: Step-reference and sinusoidal tracking experiments are executed through the HIL loop with the STM32 running the production firmware. Force tracking error and settling time are computed from ROS 2 bag data and compared against the H2 thresholds (settling time < 500 ms, error $< 15\%$).

H3 — HIL fidelity: Joint trajectories produced by the HIL configuration are compared to those from a pure-software reference simulation under identical input sequences. Root mean square deviation must remain below 2° and USB serial round-trip latency below 10 ms, confirming the embedded implementation is ready for physical hardware integration in Phase 2.

All data are logged via ROS 2 bag files and processed offline in MATLAB/Python. Results are reported as mean $\pm$ standard deviation across trials.

Stage deliverable: HIL validation dataset and statistical analysis report confirming H1, H2, and H3.

8. SCHEDULE

The development of PRANK Phase 1 is structured into the four stages described in the Materials and Methods chapter. The project timeline spans **7 months** (2026), with parallel execution of independent workstreams where feasible to maximise efficiency within the available time, as illustrated in Figure 8.1.

The schedule is organised as follows:

- **Stage 1 — Mechanical Design and CAD Modelling (Months 1–2):** 3D design in SolidWorks, structural FEA, kinematic workspace verification via SolidWorks Motion, and URDF/SDF export for simulation import.
- **Stage 2 — Electronic Architecture Design (Months 2–3):** Component selection (BLDC motors, encoders, motor drivers, six-axis load cell, STM32 board), schematic capture, PCB layout, bill of materials, and dual-core firmware architecture specification (FreeRTOS, CAN bus, ADC). The STM32 development board is configured and the firmware skeleton deployed during this stage.
- **Stage 3 — Embedded Control Implementation (Months 3–5):** C/C++ implementation of the admittance control law on the STM32 (passive/assistive/resistive modes), FreeRTOS task partitioning, safety monitors, and USB serial communication with the ROS 2 simulation host. The ROS 2/Gazebo simulation environment is configured with the URDF model and the HIL bridge node developed in parallel.
- **Stage 4 — HIL Validation and Performance Evaluation (Months 5–7):** Full HIL loop commissioning, step-reference and sinusoidal tracking experiments, H1 workspace verification in simulation, H2 controller performance evaluation, and H3 HIL fidelity assessment. Statistical analysis in MATLAB/Python and final report writing.

Continuous **documentation** of design decisions, experimental results, and the final pre-project report runs across all seven months. The milestone *HIL Loop Ready* is placed at Month 5, marking the point at which the STM32 firmware and the ROS 2/Gazebo simulation are fully integrated and the validation experiments can begin.

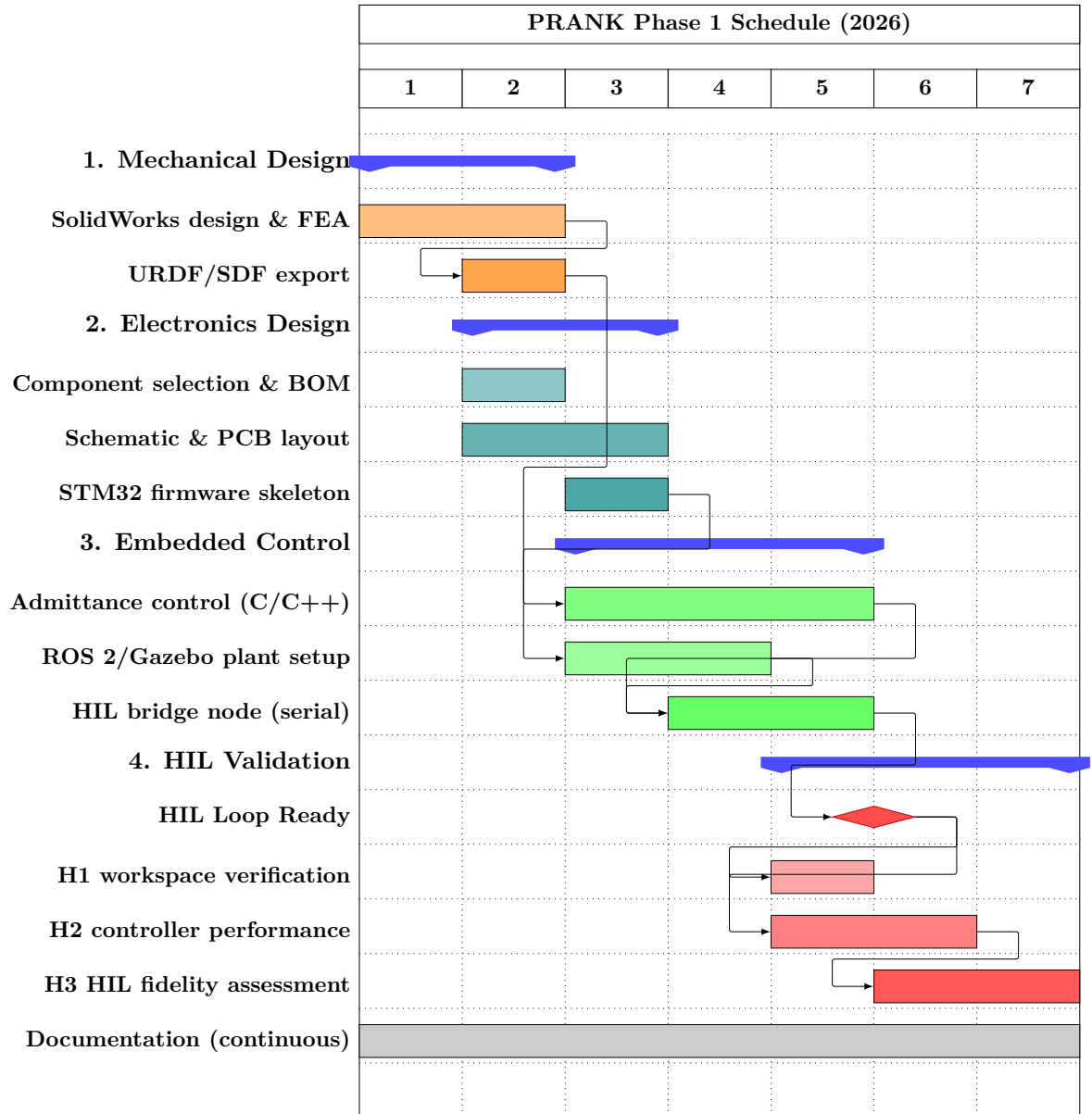


Figure 8.1: Seven-month Phase 1 schedule for PRANK. Colours indicate stage: orange (mechanical design), teal (electronics design), green (embedded control and simulation), red (HIL validation), grey (documentation).

9. PROJECT RESOURCES

This chapter outlines the technical requirements and estimated costs for PRANK Phase 1, focused on design and computational simulation. Since no physical fabrication is carried out in this phase, the budget reflects software tools, development hardware for hardware-in-the-loop testing, and engineering work hours.

Table 9.1: Resource and Cost Estimation for PRANK Phase 1 (Design & Simulation)

Category	Description	Cost (USD)
STM32 Dev Board	STM32H7/F7 dual-core kit for HIL firmware validation	60
USB Serial Interface	USB-to-UART/CAN adapter for ROS 2 HIL bridge	20
Electronic Design Tools	KiCad (free) — schematic and PCB layout	0
Software Licenses	SolidWorks, MATLAB (university); ROS 2, Gazebo, Ubuntu (free)	100
Miscellaneous	Cables, adapters, lab consumables	30
Engineering Hours	300 h $\times$ \$15/hr — CAD design, electronics spec, firmware, HIL validation	4,500
Contingency (10%)	Unexpected software issues, additional hardware	471
Total		5,181

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ANNEXES